

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 9: Ground De-Icing, Anti-Icing & Holdover Times (HOT)

Syllabus Source: CAE Oxford (pp. 311-346) | Pass Target: >= 90% AviationExam

1. THE CLEAN AIRCRAFT CONCEPT & STANDARD FLUIDS

Under EASA CAT.OP.MPA.250, no aircraft may commence take-off unless all critical surfaces (wings, control surfaces, engine inlets, pitot tubes) are completely free of frost, ice, slush, or snow:

ISO / SAE Fluid Type	Viscosity, Thickener & Dye Color	Aerodynamic & Operational Characteristics
ISO / SAE Fluid Type	Viscosity, Thickener & Dye Color	Aerodynamic & Operational Characteristics
Type I (De-icing)	Unthickened, low viscosity. DYED ORANGE.	Applied hot (60°C) to remove ice/snow. Minimal Holdover Time (HOT: 5 to 15 min).
Type II (Anti-icing)	Thickened pseudoplastic. DYED YELLOW / CLEAR.	Contains polymeric thickeners that absorb freezing moisture. Minimum rotation speed: 100 kt. HOT: 30 to 45 min.
Type III (Commuter)	Thickened, intermediate viscosity. DYED BRIGHT GREEN.	Designed for commuter turboprops with low rotation speed (Vr < 100 kt).
Type IV (Advanced Anti-icing)	Highly thickened pseudoplastic. DYED EMERALD GREEN.	Provides LONGEST HOLDOVER TIME. Shears off cleanly during take-off roll at Vr >= 100 kt.

2. HOLDOVER TIME (HOT) RULES

Holdover Time is the estimated time anti-icing fluid will prevent frost/ice forming on treated aircraft surfaces:

- Exact Start of HOT: Holdover Time BEGINS AT THE EXACT MOMENT THE FINAL ANTI-ICING APPLICATION COMMENCES! (Not when treatment finishes!).
- Pre-Take-Off Contamination Check: If HOT expires prior to take-off, the aircraft must return for complete de-icing/anti-icing re-treatment, unless an external pre-take-off contamination inspection confirms surfaces remain 100% clean.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Holdover Time (HOT) Management

Clean Aircraft Concept Rules: Step 1: When final Type IV anti-icing fluid application begins: - START STOPWATCH! Holdover Time begins at START of final application, NOT when completed. Step 2: Check HOT table for fluid type, outside temperature, and precipitation type (snow, freezing fog, rain). Step 3: If taxi delay approaches end of HOT: - Inside HOT: Surfaces glossy and fluid -> Cleared for takeoff. - Expired HOT: Perform pre-takeoff contamination check. If surfaces lose gloss or show snow adhesion -> RETURN FOR RE-DE-ICING!

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Exactly when does the HOLDOVER TIME (HOT) of an anti-icing fluid treatment begin?

- [A]: When the de-icing truck leaves the aircraft
- [B]: At the START of the final anti-icing application
- [C]: When take-off clearance is received
- [D]: At brake release

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. By international ICAO and EASA definition, holdover time begins at the exact start of the final anti-icing application step.

Q2: What visual color dye is standard for SAE TYPE IV advanced anti-icing fluid?

- [A]: Orange
- [B]: EMERALD GREEN
- [C]: Yellow
- [D]: Blue

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Type I is orange; Type II is straw/yellow; Type III is bright green; Type IV is emerald green.

Q3: What happens to a thickened pseudoplastic anti-icing fluid (Type II / IV) during the aircraft's take-off ground roll?

- [A]: It freezes solid
- [B]: Under aerodynamic shear stress at speeds above 100 kt, its viscosity collapses and it SHEARS COMPLETELY OFF THE WING, leaving a clean aerodynamic surface
- [C]: It remains permanently on the wing
- [D]: It dissolves fuel

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Pseudoplastic fluids thin under shear stress, flowing completely off the wing at high speed during rotation to avoid reducing wing lift.